

Continual Learning and Affective State Identification Across World Languages

Anonymous ACL submission

Abstract

1 Introduction

ND:[Prior emotion detection work/why is emotion understanding important] Historically, research on sentiment and emotion detection has relied on small, predetermined label sets (e.g., Ekman (1984, 1992), Plutchik (1980)) that are applied universally across varying linguistic and cultural backgrounds. While recent work has considered larger emotion label sets (Wang et al., 2020; Mohammad and Kiritchenko, 2015; Lian et al., 2024), few studies have considered open-vocabulary settings grounded in natural language emotion expression.

ND:[Why affective states/open vocabulary] The predetermined label sets used in traditional emotion detection datasets have several limitations: (1) emotion taxonomies can struggle to capture nuanced emotions such as *grief* and (2) emotions are typically labeled by third-party annotators who bring their own cultural and (3) linguistic perspectives to the annotation task. To address these issues, we instead derive labels from natural language, avoiding annotator biases and allowing variation across speakers.

ND:[Summary of background on emotion variation] Perhaps most importantly, traditional emotion taxonomies may be applied universally, failing to give attention to variation in cognitive understanding of emotion terms, e.g., across cultural and linguistic boundaries (Jackson et al., 2019). Work has shown, for example, that *frustration* in English is closer to *anger* while in different European languages, it can denote other emotional experiences (Soriano and Ogarkova, 2025). Alternatively, work has shown that such linguistic and cultural variation does not extend to language models, finding that emotions like *pride* are primarily interpreted from a Western lens (Havalдар et al.,

2023). Accordingly, it is vital to consider this variation when evaluating LLMs’ understanding of emotions across languages.

In this work, we focus on evaluating multilingual LLMs’ understanding of affective states—fine-grained and nuanced labels of emotions and feelings—across a variety of languages. We build on prior work (Deas et al., 2024) to collect a large-scale dataset of personal narratives with self-reported affective states, considering a broader set of linguistic patterns, sources, and languages. We then leverage this dataset toward two lines of work toward characterizing models understanding of emotions across languages: 1) studying training dynamics in this domain and 2) studying models’ representations of affective state labels. We outline our contributions as follows:

1. We curate a **large-scale, native-speaker validated dataset for affective state identification**, ND:[Need a name], covering **eight languages** with varying resourcedness and typologies.
2. We benchmark multilingual LLMs and study **cross-lingual training dynamics** on this task, finding that **TODO: [...Summarize trends]**
3. We use the collected benchmark to study **models’ internal representations** of fine-grained affective states and **how those representations are impacted by training**. We find that **TODO: [Findings]**

2 Background and Related Work.

3 Data

ND:[(Nick) What is the goal of the data detection (i.e., language coverage, native speaker personal narratives, diverse linguistic contexts.)]

ND:[Explicitly distinguish from A) other emotion datasets, B) MASIVE, and C) open-vocabulary annotations]

3.1 Languages

Language	Code	Family	Branch	Tier
English	eng	Indo-European	Germanic	5
Spanish	spa	Indo-European	Italic	5
French	fra	Indo-European	Italic	5
Romanian	ron	Indo-European	Italic	3
Persian	pes	Indo-European	Indo-Iranian	4
Hindi	hin	Indo-European	Indo-Iranian	4
Mandarin	cmn	Sino-Tibetan	Chinese	5
Indonesian	ind	Austronesian	Malayo-Polynesian	3

Table 1: Included languages with language family information, ISO 639-3 code and resource tiers (Joshi et al., 2020). Unless specified in the ISO code, the region or dialect is non-specific. ND:[Should add scripts here as well]

3.2 Data Sources

Source	Lang.	License
OSCAR	All	CC0
CC-100	All	CC0
Reddit	All	Public API
YouTube	All	Public API
HmBlogs	pes	Research
OpenSubtitles	hin	CC BY-SA
EmotionTalk	cmn	CC BY-NC-SA 4.0
NSMC ¹	kor	CC BY 2.0

Table 2: ND:[@Parmida: Table with sources across languages (like SemEval)]PS:[need to update ron/hin/kor/cmn data but placeholder for now]

ND:[@Parmida, Quick description/citation (if needed) of the broad sources we used (e.g., we can just group prior emotion datasets under one heading). Can point to an Appendix with specifics about preprocessing, special considerations, etc.] We collect formal and informal text with expressions of affective states from sources that fall into web-scale crawls, community platforms, and curated language-specific corpora; Table 2 lists the assignment of sources to languages. These source classes capture different styles of expression within each language, ranging from edited web prose to spontaneous speech.

3.3 Data Collection

Bootstrapping ND:[(Nick) Briefly describe/cite MASIVE]

Prior work (Deas et al., 2024) introduced a data collection approach for gathering open-ended affective states instead of fixed emotion classes. We extended the methodology by implementing

Lang	Context	Label
eng	...I feel __ in myself that I did not report him to the police or have a pepper spray...	dissapointed
spa	...Que no es como si estuviera __ por conseguir novia pero preferiría amigos...	desesperado desperate
pes	احساس __ و ناامیدی داشتم...	غم grief
hin	bahut __ hoon aaj yaar...	sad
ron	Sunt __ să-mi vând toate lucrurile pt bani de plecare...	disperat desperate

Table 3: Abbreviated context and label examples. PS:[I will update this later to add more languages and the somatic-idiom examples with a new column for pattern. I think we should also add the translation of the non english one as well]

Language	# Train	# Val	# Test	# Aff. States	Avg. Len.
English	~93	~10k		1,627	~311
Spanish	~30			1,002	~241
Persian	~				
Hindi	~				
Romanian	~				
French	~				
Korean	~				
Mandarin	~				

Table 4: Summary data statistics for each language including total size (# Docs), number of unique labels (# Aff. States), and text lengths are measured in mT5 tokens (Avg. Len.).

a shared bootstrapping pipeline: (1) per language seed affective states from emotion lexica or translated emotion categories, (2) extract first-person contexts via conjunction patterns (e.g., "I feel X and Y" where X is the seed) from corpora, (3) add newly discovered candidates to seed list and iterate, for a fixed number of rounds, and (4) validate candidates through LLM and human annotation. For languages that express emotion through somatic idioms, we extended the pipeline by extracting context view somatic patterns: e.g., Persian (del-am X shod), where the body part *del* (heart) in relation to X is used to express an emotion, and the idiom as a whole expresses the affective state (Sharifian, 2008; Rezai and Samenian, 2023). In addition to the 8 listed languages, we capture transliterated and code-switched text, e.g., Hindi-English (*Hinglish*). After extraction, every affective state is masked (both seed and discovered conjuncts; somatic idioms masked as the whole phrase), examples are deduplicated, and split into train/val/test as seen in Table 4.

3.4 Language-Specific Methods

TODO: [All: Temporary section for now so we have all of the details for each language that we can merge/organize between main paper and ap-

pendix. Please include details like those that are skeletoned out for English.]

English. TODO: [Where did you get your initial seed set from? (i.e., did you start with Ekman or another set?)]

TODO: [What templates did you use?]

- Template 1 (rough translation)
- Template 2 (rough translation)
- ...

TODO: [Any other necessary changes for your data? (i.e., did bootstrapping not work? Anything else you did differently/additionally? Why?]

Spanish.

French. Initial seed from reddit esp. r/besoindeparler; template is je me sens X et Y, Y is new candidate state. "heureux", "heureuse", "triste", "furieux", "furieuse", "effrayé", "effrayée", "dégoûté", "dégoûtée", "surpris", "surprise"

Romanian.

Persian.

- **Initial Seed Construction.** 12 initial seeds: the six Ekman categories (Ekman, 1992) in both noun form (ترس, خشم, غم, شادی, نفرت, تعجب) and adjective form (خوشحال, متنفر, متعجب, ترسان, خشمگین, غمگین). Including both forms is required because of how Persian handles emotional predication. Specifically, Persian relies heavily on light-verb complex predicates; the frame "I feel" («احساس X می‌کنم») requires a nominal argument (Karimi-Doostan, 2005; Folli et al., 2005). While, adjectival states must predicate through a first-person copula or inchoative verb like «X شدم» (Mostafavi, 2015).

- **Template used for data extraction.**

- **Noun frame.** «احساس X و Y می‌کنم» (lit. "I make the feeling of X and Y" ≈ "I feel X and Y")
- **Adjective frame.** «X و Y شدم» ("I became X and Y")
- **Somatic-idiom frame.** «دلم X شد» / «دلم گرفت» ("my heart became X" / "my heart seized" ≈ "I felt down"). Body part + 1sg possessive enclitic م + verb. Here, the enclitic satisfies the required

first-person constraint. Because these expressions are non-compositional, the idiom is masked as a single, contiguous phrase (Sharifian, 2008; Rezai and Same-nian, 2023).

Hindi.

Mandarin.

- **Initial seed construction.** Started with Chinese translations of the six Ekman basic emotion categories (Ekman, 1992). Initial seed set included common Mandarin forms such as 开心/高兴 (happy), 难过/伤心 (sad), 生气/愤怒 (angry), 害怕/恐惧 (afraid), 惊讶 (surprised), and 恶心 (disgusted).
- **Templates used for data extraction.** We used a variety of Mandarin templates such as self-state and conjunctive/contrastive templates. These were used in different stages of the affective state extraction pipeline. Example templates include:
 - Self-state: 我很 X (I am very X)
 - Self-state: 我有点 X (I am a little X)
 - Self-state: 我觉得 X / 我感觉 X (I feel X)
 - Conjunctive: 我又 X 又 Y (I am both X and Y)
 - Contrastive: 我很 X 但也很 Y (I am very X but also very Y)
 - 这让我很 X (This makes/made me very X)
- **Other language-specific changes.** Bootstrapping worked for Mandarin, but several additional cleanup steps were needed. First, we normalized text toward Simplified Chinese to reduce duplication between Traditional and Simplified variants. Second, for YouTube/Filmot transcripts, we restored punctuation before segmentation because many subtitle files were under-punctuated. Third, we added an extra label-level audit before dataset creation because some Chinese strings that look affective in isolation are often non-affective within specific contexts. For example, 希望 can mean "hope" or "want," 感谢 is often a thanking formula, 安全 often refers to physical/security safety, and 对不起 is often an apology formula. We sampled

218 matched examples for high-risk labels and re-
219 moved labels judged too noisy before creating
220 masked examples.

221 **Indonesian.**

222 **3.5 LLM Filtering**

223 **3.6 Data Validation**

224 ND:[Human annotations]

225 **4 Experimental Methods**

226 **4.1 Models**

227 ND:[Qwen3.5, gpt-oss?]

228 **4.2 Metrics**

229 **Top-k Accuracy.**

230 **Top-k Similarity.**

231 **4.3 Zero-Shot Evaluation**

ND:[...]

Figure 1: ND:[Prompt used for evaluation]

232 **4.4 Training Dynamics**

233 ND:[@Both: For now, quick details on how
234 the task was structured for instruct-tuned mod-
235 els/how you did finetuning for the group pre-
236 sentation experiments. Don't need to in-
237 clude specific hyperparameter values like max
238 length or lora params (if used) unless they
239 are relevant to interpreting the results, just
240 higher-level mentions like "Constrained gen-
241 eration length" and reference Appendix D
242 with the specifics.]

243 **4.5 Activation Space Structure**

244 **5 Results**

245 **5.1 Zero-Shot Evaluation**

246 **5.2 Finetuning Evaluation**

247 ND:[What all do we want to show in the main
248 paper here?]

5.3

6 Conclusion

Limitations

Ethics Statement

References

- Nicholas Deas, Elsbeth Turcan, Iván Pérez Mejía, and Kathleen McKeown. 2024. [MASIVE: Open-ended affective state identification in English and Spanish](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 20467–20485, Miami, Florida, USA. Association for Computational Linguistics.
- Paul Ekman. 1984. Expression and the nature of emotion. *Approaches to emotion*, 3(19):344.
- Paul Ekman. 1992. [An argument for basic emotions](#). *Cognition and Emotion*, 6(3–4):169–200.
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- S. Ghafouri et al. 2023. Ariabert: A pre-trained persian bert model for natural language understanding. In *Proceedings of International Conference on Persian NLP*.
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- Reihaneh Iranmanesh, Saeedeh Davoudi, Pasha Abrishamchian, Ophir Frieder, and Nazli Goharian. 2026. Taraz: Persian short-answer question benchmark for cultural evaluation of language models. In *arXiv preprint arXiv:2602.22827*.

Language	Model Checkpoint	Citation	Why?
English	ND:[copy huggingface checkpoint]	ND:[copy citation]	ND:[Any reason this model is better than others? (e.g., benchmark performance, etc.)]
Spanish			
French			
Romanian			
Persian	universitytehran/PersianMind-v1.0 MehdiHosseiniMoghadam/AVA-Llama-3-V2	(Rostami et al., 2024) (Iranmanesh et al., 2026)	PersianMind (7B): Adaptation of Llama-2-7B. Has paper publication(arxiv only), vocab expanded by 10k tokens. Caveat: Older base model and TARAZ found to not be as good as other models. AVA-Llama-3 (8B): Adaptation of Llama-3-8B. SOTA Llama 3 backbone; formally evaluated on the TARAZ cultural short-answer benchmark and found to be the best open Persian finetuned one. Caveat: No standalone model paper.
Hindi			
Mandarin	Qwen/Qwen2.5-7B-Instruct	(Yang et al., 2024)	Qwen2.5-7B-Instruct: Chinese-native Qwen-family instruction-tuned model. Improved pre-training and post-training over earlier Qwen models, including scaling pretraining data to 18T tokens.
Indonesian			

Table 5: LLMs created specifically for each language. ND:[@All: Need to be 1) instruction-tuned, 2) have a 7-9B variant. Preferably a model that is relatively recent (last 3-5 years), has an attached publication in a peer-reviewed venue (i.e., not just arxiv), and is an adaptation of a Llama, Qwen, or Mistral model.]

Language	Model Name	Checkpoint	Citation	Why?
English				
Spanish				
French				
Romanian				
Persian	AriaBERT	ViraIntelligentDataMining/AriaBERT	(Ghafouri et al., 2023)	Upgraded RoBERTa-base model using a BPE tokenizer. Outperforms ParsBERT on downstream tasks by up to 3%.
Hindi				
Mandarin	BGE-large-zh-v1.5	BAAI/bge-large-zh-v1.5	(Xiao et al., 2023)	BERT/feature-extraction based, 1024-dimensional, and evaluated on C-MTEB, a Chinese text embedding benchmark.
Indonesian				

Table 6: Embedding models created specifically for each language. ND:[@All: should be encoder-only (i.e., bert/roberta-based)]

293	Joshua Conrad Jackson, Joseph Watts,	zabān-e fārsi: negāhi rade-šenāxti]. <i>Jour-</i>	340
294	Teague R. Henry, Johann-Mattis List,	<i>nal of Linguistics & Khorasan Dialects</i> ,	341
295	Robert Forkel, Peter J. Mucha, Simon J.	15(3):1–27. Serial No. 32, Fall 2023. In	342
296	Greenhill, Russell D. Gray, and Kristen A.	Persian.	343
297	Lindquist. 2019. Emotion semantics		
298	show both cultural variation and universal	Pedram Rostami, Ali Salemi, and Mo-	344
299	structure . <i>Science</i> , 366(6472):1517–1522.	hammad Javad Dousti. 2024. Persian-	345
300		mind: A cross-lingual persian-english	346
301	Pratik Joshi, Sebastin Santy, Amar Budhiraja,	large language model. In <i>arXiv preprint</i>	347
302	Kalika Bali, and Monojit Choudhury. 2020.	<i>arXiv:2401.06466</i> .	348
303	The state and fate of linguistic diversity and		
304	inclusion in the NLP world . In <i>Proceedings</i>	Farzad Sharifian. 2008. Conceptualizations of	349
305	<i>of the 58th Annual Meeting of the Associa-</i>	del ‘heart-stomach’ in Persian . In Farzad	350
306	<i>tion for Computational Linguistics</i> , pages	Sharifian, René Dirven, Ning Yu, and Su-	351
307	6282–6293, Online. Association for Com-	sanne Niemeier, editors, <i>Culture, Body,</i>	352
	putational Linguistics.	<i>and Language: Conceptualizations of In-</i>	353
308		<i>ternal Body Organs across Cultures and</i>	354
309	Gholamhossein Karimi-Doostan. 2005.	<i>Languages</i> , pages 247–265. De Gruyter	355
310	Light verbs and structural case . <i>Lingua</i> ,	Mouton, Berlin and New York.	356
	115(12):1737–1756.		
311		Cristina Soriano and Anna Ogarkova. 2025.	357
312	Zheng Lian, Haiyang Sun, Licai Sun, Zhuo-	The meaning of ‘frustration’ across lan-	358
313	fan Wen, Siyuan Zhang, Shun Chen, Hao	guages . <i>Language and Cognition</i> , 17.	359
314	Gu, Jinming Zhao, Ziyang Ma, Xie Chen,		
315	Jiangyan Yi, Rui Liu, Kele Xu, Bin Liu,	Zhaoxia Wang, Seng-Beng Ho, and Erik	360
316	Erik Cambria, Guoying Zhao, Björn W.	Cambria. 2020. A review of emotion	361
317	Schuller, and Jianhua Tao. 2024. Mer	sensing: categorization models and algo-	362
318	2024: Semi-supervised learning, noise ro-	rithms . <i>Multimedia Tools and Applications</i> ,	363
319	bustness, and open-vocabulary multimodal	79(47–48):35553–35582.	364
320	emotion recognition . In <i>Proceedings of</i>		
321	<i>the 2nd International Workshop on Multi-</i>	Shitao Xiao, Zheng Liu, Peitian Zhang, and	365
322	<i>modal and Responsible Affective Comput-</i>	Niklas Muennighoff. 2023. C-pack: Pack-	366
323	<i>ing</i> , MRAC ’24, page 41–48, New York,	aged resources to advance general chinese	367
324	NY, USA. Association for Computing Ma-	embedding . <i>Preprint</i> , arXiv:2309.07597.	368
	chinery.		
325		An Yang, Baosong Yang, Beichen Zhang,	369
326	Saif M. Mohammad and Svetlana Kir-	Binyuan Hui, Bo Zheng, Bowen Yu,	370
327	itchenko. 2015. Using hashtags to capture	Chengyuan Li, Dayiheng Liu, Fei Huang,	371
328	fine emotion categories from tweets . <i>Com-</i>	Haoran Wei, et al. 2024. Qwen2.5 technical	372
	<i>putational Intelligence</i> , 31(2):301–326.	report. <i>arXiv preprint arXiv:2412.15115</i> .	373
329			
330	Faezeh Saadat Mostafavi. 2015. Predica-	A Model Details	374
331	tive clauses in today Persian: A typo-		
332	logical analysis . <i>Asian Social Science</i> ,	B Data Source Details	375
	11(15):104–110.		
333		Sources	376
334	Robert Plutchik. 1980. A general psychoevo-		
335	lutionary theory of emotion. In <i>Theories of</i>	Licenses	377
	<i>emotion</i> , pages 3–33. Elsevier.		
336		C Query Template Details	378
337	Vali Rezai and Tahereh Samenian. 2023.		
338	Body-part terms and the expression of emo-		
339	tions in Persian: A typological perspective		379
	[nām-e andāmhā va bayān-e eḥsāsāt dar		

	fa	ro	hi	fr
Seeds (initial → final)	12 → TODO	375 → 524	64 → 133	11 → 298
Frames	3 families	20 patterns	3 stages	2 templates
Conjunction bootstrapping	(3 iter)	(failed)		(5 iter)
Somatic channel	(template)		(detection)	
Masked-LM expansion	(validation)		(MuRIL)	
Distributional patterns				
Sources	3	8	7	3
LLM filter (0–3, open)	TODO		partial	partial
Human validation	TODO	($\kappa_w=.65$)	planned	
Examples / labels	TODO	73,427 / 910	22,432 / 120	76,927 / 298

Table 7: Per-language breakdown TODO caption text.

D Generation and Finetuning Details

ND:[We can include the specific values of hyperparameters and such here rather than main paper.]